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Review

Understanding Creativity Across Brains and Machines Through the GEEC Framework

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Abstract

Creativityis often viewed as one of the defining hallmarks of intelligence, yet the mechanisms underlying creative behavior remain poorly understood. This question has become increasingly important as artificial intelligence systems demonstrate remarkable abilities to generate novel text, images, music, and code, raising the possibility of more general forms of machine creativity and, ultimately, Artificial General Intelligence (AGI)—the capacity for flexible intelligence across diverse tasks and domains. Although creativity is frequently treated as a singular capability, evidence from cognitive neuroscience suggests that it instead emerges from interactions among multiple brain systems responsible for generative, evaluative, and exploratory processes regulated by coordination. In this paper, we propose the Generative–Evaluative–Exploratory–Coordination (GEEC) framework, a neuroscience-inspired functional framework that conceptualizes creativity as an emergent property of three primary creative processes—generation, evaluation, and exploration—whose interactions are dynamically regulated by a higher-order coordination process. Generative processes produce novel ideas through imagination and associative recombination; evaluative processes assess their utility and coherence; exploratory processes drive curiosity and learning under uncertainty; and coordination is conceptualized as a meta-process that regulates transitions among the other components. We review the neural mechanisms underlying these functions and examine how they have informed modern computational approaches, including generative models, reinforcement learning, meta-learning, neuro-symbolic systems, and cognitive architectures. Rather than viewing neuroscience as a roadmap to AGI, we argue that it provides a source of hypotheses about the computational principles that support creative behavior. The GEEC framework emphasizes functional correspondences rather than mechanistic equivalence and proposes that creativity is inherently multicomponent, emerging from dynamic interactions among generative, evaluative, and exploratory processes regulated through coordination. By synthesizing perspectives from neuroscience, cognitive science, and artificial intelligence, GEEC offers a unifying framework for understanding creativity across brains and machines.

Keywords: creativity; artificial general intelligence; cognitive neuroscience; generative AI; reinforcement learning; salience network

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1. Introduction

Creativity is widely regarded as one of the defining hallmarks of intelligence. It enables individuals to generate ideas, solutions, and artifacts that are not only novel, but also

useful, meaningful, or surprising within a particular context [1]. In humans, creativity underlies scientific discovery, artistic expression, flexible problem solving, and adaptation to uncertain environments. As artificial intelligence systems become increasingly capable of producing novel text, images, music, code, and other multimodal content, understanding the computational principles that support creative behavior has become an increasingly important challenge. If Artificial General Intelligence (AGI) is understood as the capacity for flexible, cross-domain intelligence rather than narrow task performance, then creativity provides an important domain for examining capabilities relevant to general intelligence, including adaptation to unfamiliar situations, transfer across domains, and the generation of solutions that extend beyond prior experience.

Historically, attempts to explain creativity have alternated between viewing it as a singular faculty and conceptualizing it as an emergent property arising from interactions among multiple cognitive processes. Early psychometric approaches emphasized stable individual differences in creative ability [2], whereas contemporary cognitive neuroscience increasingly views creativity as the product of dynamic interactions among distributed brain systems supporting imagination, executive control, memory, and attention [3,4]. Rather than localizing creativity to a single brain region or cognitive faculty, this perspective suggests that creative behavior emerges through the coordinated interaction of multiple functional processes.

Recent advances in artificial intelligence have made this question especially timely. Large language models, diffusion models, and other generative systems can now produce outputs that often appear creative. These advances have renewed questions about the relationship between novelty, value, intentionality, autonomy, and genuine creativity [1,5,6]. Although these systems demonstrate remarkable generative capabilities, it remains unclear whether generation alone is sufficient for creativity or whether creative intelligence requires additional mechanisms for evaluating ideas, exploring alternatives, and regulating behavior across changing goals and contexts.

Motivated by these developments, we propose the **Generative–Evaluative–Exploratory–Coordination (GEEC)** framework, a neuroscience-inspired functional framework that characterizes creativity as emerging from three primary functional processes—generation, evaluation, and exploration—whose interactions are dynamically regulated by coordination. Generative processes construct candidate ideas through imagination, simulation, and associative recombination. Evaluative processes determine whether those ideas are useful, coherent, or appropriate for current goals and constraints. Exploratory processes promote curiosity, novelty seeking, and learning under uncertainty, thereby enabling creative systems to discover new possibilities beyond their existing knowledge. Coordination, however, differs fundamentally from these processes. Rather than representing an additional source of creative behavior, it refers to higher-order regulation of when and how generative, evaluative, and exploratory processes are deployed. Viewed another way, generation, evaluation, and exploration describe the principal functional operations of creative cognition, whereas coordination governs their dynamic interaction across changing goals, contexts, and timescales.

Figure 1 summarizes the GEEC framework. Importantly, GEEC should be understood as a functional framework rather than a mechanistic model. Its purpose is not to suggest one-to-one correspondences between particular brain regions and AI architectures, but to identify common computational challenges that both biological and artificial systems must solve in order to exhibit flexible and adaptive creative behavior. From this perspective, neuroscience serves not as a blueprint for AGI, but as a source of hypotheses regarding the functional organization of creative systems. GEEC does not assume that creativity is either necessary or sufficient for AGI. We argue that the capacities implicated in creative cognition—generating alternatives, evaluating them against goals, exploring uncertain

solution spaces, and adaptively coordinating these processes—overlap with computational challenges faced by systems designed to generalize flexibly across domains. Creativity, therefore, provides a useful test case for studying capabilities relevant to general intelligence rather than a defining criterion for AGI itself.

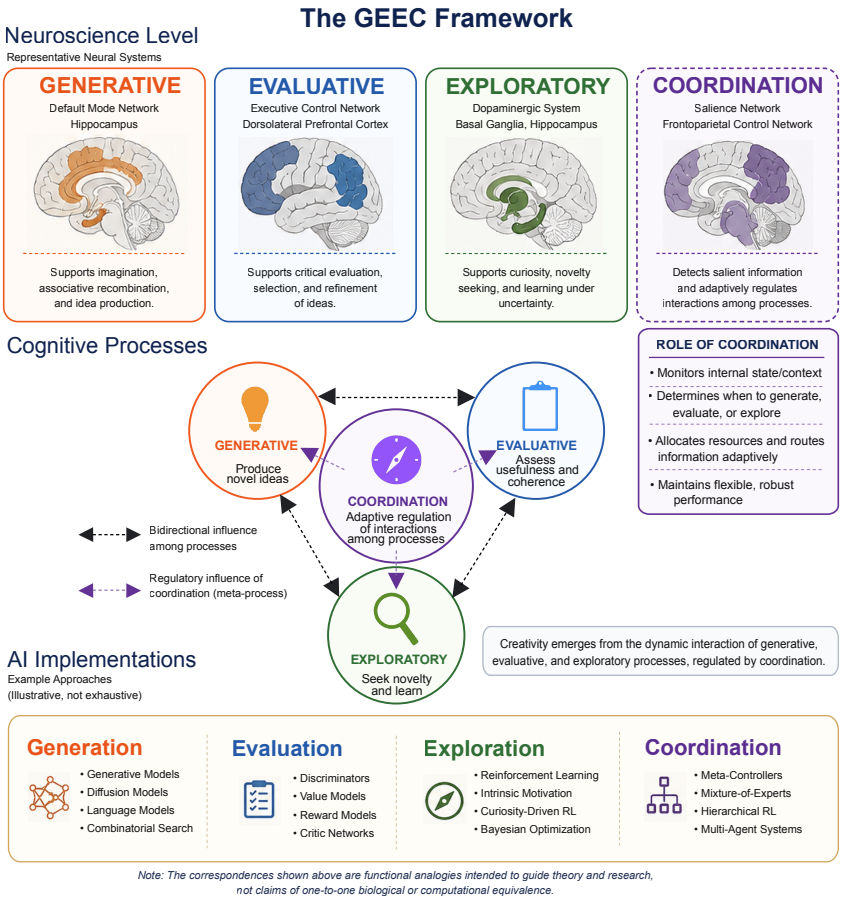


Figure 1. The GEEC framework proposes that creativity emerges through the dynamic interaction of three complementary cognitive processes—generation, evaluation, and exploration—whose interactions are adaptively regulated by coordination. The (**top**) panel summarizes representative neural systems associated with each functional component based on the current neuroscience literature. The (**central**) panel illustrates the reciprocal interactions among the three primary creative processes and the regulatory role of coordination as a meta-process that monitors context, allocates resources, and adaptively guides transitions among them. The (**lower**) panel presents illustrative examples of corresponding computational approaches in artificial intelligence. These correspondences are intended as functional analogies that highlight shared computational challenges rather than one-to-one biological or mechanistic equivalences. Together, the framework provides a multilevel organization that links neural systems, cognitive functions, computational principles, and candidate AI implementations while generating testable hypotheses about creativity in both biological and artificial systems.

GEEC builds upon several influential theories of creativity and should be viewed as an integrative framework rather than a replacement for existing models. Its contribution lies not in proposing entirely new creative processes, but in providing a common functional vocabulary linking cognitive functions, associated neural systems, computational principles, contemporary AI architectures, and approaches to creativity assessment. The relationship between GEEC and existing frameworks is considered in greater detail below.

More broadly, GEEC has implications for how creativity is evaluated. Creative performance depends not only on the ability to generate novel ideas, but also on the capacity to

evaluate those ideas, explore new possibilities, and effectively coordinate these processes over time. This perspective motivates a multicomponent approach to creativity assessment and generates testable hypotheses regarding the neural, cognitive, and computational processes that support creative behavior.

The remainder of this review first examines the neuroscience of creative cognition through the lens of the GEEC framework, then considers how these principles have informed contemporary computational models, and finally discusses how creativity may be operationalized and evaluated in both biological and artificial systems.

2. Creativity in the Human Brain

The neuroscience literature does not support a one-to-one mapping between creative functions and discrete neural structures. Instead, generation, evaluation, and exploration are associated with partially dissociable large-scale networks and neuromodulatory systems whose contributions overlap and interact. Coordination is different in kind: it refers to regulation of those interactions through mechanisms such as attention, salience detection, and cognitive flexibility. The neural systems discussed below should therefore be read as functional emphases rather than dedicated "creativity modules".

2.1. Generative Processes

Generative processes are responsible for the production of candidate ideas through imagination, mental simulation, and associative recombination. Within the GEEC framework, this function is most closely associated with the default mode network (DMN), a large-scale network including the medial prefrontal cortex, posterior cingulate cortex, precuneus, and angular gyrus [7,8].

The DMN is active during internally directed cognition, including autobiographical memory retrieval, future thinking, mind wandering, and imagination. Rather than passively reflecting ongoing mental activity, the network appears to support the construction of hypothetical scenarios through the flexible recombination of stored experiences into novel representations [9]. This capacity for episodic simulation provides a plausible neural basis for creative ideation by enabling individuals to mentally explore possibilities that extend beyond immediate sensory experience.

Consistent with this view, both neuroimaging and causal intervention studies have implicated the DMN in divergent thinking. Direct cortical stimulation of DMN regions disrupts performance on alternate-uses tasks, while intracranial recordings reveal task-dependent DMN dynamics during both mind wandering and creative ideation [10,11]. Additionally, greater functional connectivity within the DMN and stronger coupling between the DMN and executive control systems are associated with higher creative achievement and improved performance on measures of creative ideation [3].

2.2. Evaluative Processes

Creative ideas must ultimately be assessed for usefulness, coherence, and relevance. Within the GEEC framework, these evaluative functions are associated primarily with the executive control network (ECN), including the dorsolateral prefrontal cortex and posterior parietal cortex [12].

The ECN supports working memory, cognitive control, goal maintenance, and decision making. During creative cognition, these functions allow individuals to evaluate candidate ideas, suppress irrelevant associations, and refine promising solutions. This process is closely related to convergent thinking, in which multiple possibilities are narrowed to a single solution [2].

Importantly, creativity appears to depend not on the independent activity of the ECN, but on its interaction with generative processes. Highly creative individuals often show increased coupling between executive and default-mode networks, suggesting that creative thought emerges through a dynamic balance between idea generation and idea evaluation [13].

2.3. Exploratory Processes

Creativity requires more than the generation and evaluation of ideas; it also depends on the willingness to explore novel possibilities and acquire information that may prove useful in the future. Neuroscientific accounts of exploration implicate dopaminergic pathways, basal ganglia circuits, and hippocampal memory systems in curiosity, novelty seeking, and adaptive learning. GEEC draws on this literature to characterize the exploratory functions that contribute to creative cognition.

Dopamine has long been implicated in reward prediction and reinforcement learning [14]. Phasic dopaminergic signaling encodes reward prediction errors, enabling organisms to update expectations and adapt behavior in response to changing environments. These mechanisms contribute to exploratory behavior by encouraging the pursuit of uncertain but potentially rewarding outcomes and by shaping the balance between exploration and exploitation.

The basal ganglia play a complementary role in action selection and reinforcement-guided decision making, which help organisms adapt behavior based on reward history and environmental feedback [15]. More broadly, cortico-basal ganglia circuits have been proposed to support flexible learning and adaptive behavior across a wide range of cognitive tasks, providing computational mechanisms for balancing novelty seeking against the exploitation of known strategies [16].

The hippocampus may occupy a particularly important position within the GEEC framework because it contributes to both generative and exploratory aspects of creative cognition. Beyond its well-established role in episodic memory, the hippocampus supports the flexible recombination of past experiences into novel configurations, enabling individuals to imagine hypothetical scenarios, mentally simulate future events, and construct remote associations [9,17]. At the same time, hippocampal circuits contribute to novelty detection, uncertainty processing, and the exploration of cognitive and spatial environments, suggesting that memory and curiosity may be deeply intertwined [18–20].

Importantly, hippocampal memories do not merely record past experiences but can also bias future decisions by allowing value and reward information to spread across associated memories, enabling novel choices among previously unexperienced alternatives [21]. Likewise, reward prediction errors influence hippocampal memory encoding, suggesting that dopaminergic signals shape which experiences are preferentially stored and later available for imagination and creative recombination [22]. Information prediction errors—the discrepancy between expected and received information—have also been shown to enhance learning and memory, supporting the view that curiosity functions not merely as a drive for novelty but as a mechanism for prioritizing knowledge acquisition [20]. Together, these findings suggest that exploratory experiences shape not only what is learned but also what becomes available for future imagination and creative recombination.

Evidence linking dopaminergic systems to creativity comes not only from reinforcement-learning theories but also from genetic studies demonstrating associations between dopaminergic pathways and individual differences in creative achievement and divergent thinking [23,24]. These findings, together, suggest that exploratory processes do more than guide behavior toward external rewards. By promoting curiosity, novelty seeking, and the acquisition of new experiences, they expand the pool of knowledge available to generative processes and increase the likelihood of discovering information that may later contribute to creative insight. There-

fore, within the GEEC framework, exploration serves as a bridge between experience and imagination and enables creative processes to continually expand the space of possible ideas.

2.4. Coordination

Coordination occupies a different functional role within GEEC than generation, evaluation, and exploration. Rather than contributing a distinct creative operation, coordination refers to the higher-order regulation of when and how these processes are engaged and interact. Creative cognition requires continual adjustment between generating possibilities, evaluating existing ideas, and exploring alternatives during changes in goals, uncertainty, and environmental demands. Coordination, therefore, governs transitions among these cognitive modes and the allocation of resources between them. In this sense, coordination is distinct from the processes that support it: executive control may maintain goals or support evaluation, attention may prioritize information, and cognitive flexibility may enable switching, whereas coordination describes the higher-order regulation of interactions among GEEC processes.

This distinction is fundamental to the GEEC framework. Generation, evaluation, and exploration describe complementary processes that contribute directly to creative cognition, whereas coordination regulates how and when those processes are engaged. Creativity therefore emerges not from the activity of any single component in isolation but from their dynamic interaction over time. Generative processes can produce an abundance of possibilities, exploratory processes can continually seek novelty, and evaluative processes can impose increasingly stringent constraints. Left unchecked, any one of these tendencies can become maladaptive. Effective creative behavior therefore depends on mechanisms that balance imagination with constraint, novelty with utility, and exploration with exploitation.

Neuroscientifically, coordination has been linked to interactions among the salience network (SN) and frontoparietal control systems. The salience network, particularly the anterior insula and anterior cingulate cortex, is thought to facilitate the detection of behaviorally relevant information and switching between internally directed cognition supported by the default mode network and externally directed executive processes [25–28]. Complementing this role, frontoparietal control systems are thought to maintain task goals, allocate cognitive resources, and exert flexible top-down control over ongoing cognitive processing, allowing creative thought to be adaptively regulated in response to changing demands [29,30].

Within the GEEC framework, these systems are not interpreted as implementing creativity themselves, but rather as supporting the dynamic regulation of interactions among generative, evaluative, and exploratory processes. Consistent with this view, growing evidence suggests that creative performance depends not only on the activity of individual brain networks but also on the flexibility with which they interact. Individuals exhibiting greater dynamic coupling and switching among default mode, executive control, and salience networks often demonstrate superior performance on measures of divergent thinking and creative achievement [3,31]. These findings suggest that creativity depends on the adaptive orchestration of multiple cognitive systems rather than the dominance of any single network.

From a computational perspective, coordination addresses a fundamental challenge faced by both biological and artificial systems: determining which cognitive process should dominate behavior at any given moment. This requires continuously integrating information about internal goals, environmental demands, uncertainty, prior knowledge, and ongoing performance in order to regulate the balance among generation, evaluation, and exploration. Importantly, this balance is unlikely to remain fixed. Creative cognition often requires rapid transitions between divergent idea generation, critical evaluation, and exploratory information seeking as problems evolve. Within the GEEC framework, co-

ordination therefore represents not simply network switching, but the dynamic regulation of interactions among multiple creative processes across spatial and temporal scales.

2.5. GEEC Framework in Context

Taken together, these findings suggest that creativity is not localized to a single brain region or cognitive process. Rather, creative cognition emerges from interactions among processes responsible for generating candidate ideas, evaluating their utility, exploring novel possibilities, and coordinating these processes over time.

The individual components of GEEC have substantial precedents in cognitive theories of creativity, network neuroscience, and computational models of learning. GEEC is therefore not intended to replace these frameworks or to suggest that generation, evaluation, exploration, or the importance of coordinating these processes are themselves novel. Instead, GEEC is best understood as an integrative functional framework rather than a new theory of creativity in the sense of proposing previously unidentified cognitive mechanisms. Its contribution lies in specifying relationships among these processes across levels of analysis and in generating predictions about how differences in their relative engagement and coordination may give rise to similar or divergent creative outcomes. Table 1 summarizes the relationship between GEEC and three especially relevant frameworks.

Table 1. Relationship between GEEC and representative theories of creativity. Rather than replacing these models, GEEC integrates their contributions within a multilevel functional framework spanning neural systems, cognition, computation, artificial intelligence, and creativity assessment.

Framework	Primary Level	Primary Contribution	GEEC Extension
Geneplore [32]	Cognitive processes	Iterative cycle of idea generation and exploration.	Explicitly distinguishes evaluation from exploration, introduces coordination as a regulatory meta-process, and connects cognition to neuroscience and AI.
Dynamic creativity networks [3,13]	Large-scale brain networks	Creative cognition emerges from interactions among default mode and executive control systems.	Extends network interactions to computational functions, AI implementations, and creativity assessment.
Triple Network Model [27]	Network neuroscience	Describes interactions among the default mode, executive control, and salience networks.	Interprets these networks in terms of functional roles while incorporating exploratory processes (dopamine, basal ganglia, hippocampus) and computational analogues.
GEEC	Cross-level framework	Integrates generation, evaluation, exploration, and coordination.	Provides a common functional framework linking neuroscience, cognition, computational principles, AI architectures, and creativity measurement.

The Geneplore model describes creative cognition as an iterative cycle in which generative processes produce "preinventive" structures that are subsequently explored, interpreted, and elaborated [32]. This model established an influential process-oriented account of creativity and anticipated GEEC's emphasis on interactions rather than a singular creative faculty. GEEC retains this iterative view but separates *evaluation* from *exploration*. In GEEC, evaluation refers specifically to assessing and refining candidate ideas relative to goals, constraints, and standards of usefulness, whereas exploration refers to seeking new information, experiences, or regions of a problem space. This distinction permits the two processes to vary independently: a system may explore broadly but evaluate poorly, or evaluate rigorously while searching too narrowly. GEEC additionally makes coordination explicit as the meta-process that regulates transitions among generation, evaluation, and exploration.

Network-neuroscience accounts provide a complementary description at the level of large-scale brain organization. Research on creative cognition has emphasized dynamic cooperation between the default mode and executive control networks, with generative and spontaneous processes interacting with goal-directed control during creative idea production [3,4,13]. Menon's triple-network model likewise describes interactions among the default mode, central executive, and salience networks, with the salience network contributing to the detection of relevant information and switching between internally and externally oriented modes of processing [26,27]. These network models provide an important neuroscientific context for GEEC. GEEC builds on this work by considering how such network interactions relate to broader functional processes that can also be examined at cognitive and computational levels. It also incorporates exploratory mechanisms involving dopaminergic, basal ganglia, and hippocampal systems that are less central to DMN–ECN–SN accounts. Importantly, coordination within GEEC is not proposed as a fourth creative process equivalent to generation, evaluation, and exploration. Rather, it represents a higher-order regulatory function through which these processes are dynamically engaged and balanced according to changing goals and environmental demands. This distinction allows GEEC to incorporate mechanisms emphasized by executive-control and network-neuroscience accounts without equating those mechanisms with creativity itself.

GEEC also differs from these earlier frameworks in its intended scope. It links cognitive and neural accounts of creativity to contemporary artificial systems without assuming that brains and machines share mechanisms. Generative models, evaluative and reasoning systems, curiosity-driven learning, and architectural control mechanisms are treated as possible implementations of common functional challenges rather than artificial replicas of particular brain networks. The same organization is then carried into assessment, where generation, evaluation, and exploration can be operationalized separately and related to creative outcomes, while coordination is evaluated in terms of how effectively these processes are regulated and integrated.

From an AI perspective, this higher-order role distinguishes GEEC coordination from fixed routing among predefined modules. Coordination is context-sensitive. Rather than following a fixed schedule, it determines when to generate, when to evaluate, and when to explore based on the current state of the task. This includes situations in which competing objectives must be reprioritized. From this perspective, cross-domain adaptation requires not only reusable computational modules, but also the ability to reinterpret what constitutes a satisfactory solution under the goals and constraints of a new domain. Within GEEC, this capacity arises through coordination: rather than requiring entirely new generative or evaluative mechanisms for each domain, the coordination process can adjust the balance among existing processes to match the demands of a new task or domain. A system may shift from exploration-heavy strategies suited to open-ended ideation toward evaluation-heavy strategies demanded by engineering constraints, which GEEC predicts may support adaptation across domains without requiring complete domain-specific retraining.

Accordingly, the novelty of GEEC does not lie in proposing entirely new cognitive components. It lies in providing a unified, multilevel framework that: (1) distinguishes generation, evaluation, and exploration as related but separable processes; (2) conceptualizes coordination as a higher-order regulatory process rather than a fourth equivalent faculty; (3) connects these functions to associated neural systems and computational implementations; and (4) uses the resulting structure to generate hypotheses and organize the multidimensional assessment of creativity in biological and artificial systems. GEEC is therefore best understood as an integrative framework intended to expose relationships, gaps, and testable differences that may be less apparent when cognitive, neural, computational, and measurement perspectives are considered separately.

3. From Neuroscience to Computation

The relationship between neuroscience and artificial intelligence has long been bidirectional. Early neural network models were inspired by simplified theories of biological neurons, while computational models have subsequently provided powerful frameworks for interpreting learning, memory, attention, and decision making in the brain. Reinforcement learning provides a notable example of this exchange: computational theories of reward-based learning informed interpretations of dopaminergic prediction-error signals, while discoveries in neuroscience subsequently inspired new reinforcement-learning algorithms and theories of adaptive behavior [14,15,33]. More broadly, neuroscientific discoveries have motivated advances in reinforcement learning, memory-augmented architectures, meta-learning, and attention-based systems [34–37], while computational models have continued to generate experimentally testable hypotheses about biological intelligence. Together, these reciprocal influences suggest that biological and artificial intelligence often confront similar computational challenges, even when their underlying mechanisms differ.

This reciprocal relationship also provides a useful way to think about creativity. The aim is not to reproduce the biological machinery of creative thought, but to ask whether artificial systems confront comparable functional problems: constructing possibilities, judging them against goals and constraints, searching for new information, and regulating when each of these activities should dominate. Contemporary AI systems offer partial solutions to these problems through very different computational architectures.

4. Computational Models of Creativity

How might these functions be instantiated computationally? Contemporary AI systems offer several partial answers, although no existing architecture captures the full breadth of human creativity. The sections below use GEEC as an organizing lens for comparing mechanisms that generate candidate outputs, evaluate possible solutions, explore uncertain environments, and coordinate specialized capabilities.

4.1. Generative Models

The generative component of the GEEC framework concerns a fundamental challenge faced by both biological and artificial systems: how can previously acquired knowledge be transformed into novel ideas? In humans, this process is associated with imagination, episodic simulation, and associative recombination. In artificial systems, the same challenge has motivated a wide range of generative architectures designed to learn internal representations of data and use those representations to construct new outputs.

Although these architectures differ substantially in their implementations, they share a common computational principle: creativity emerges through the exploration and recombination of learned representations. Variational autoencoders (VAEs) learn structured latent spaces that can be manipulated to generate new observations [38]. Generative adversarial networks (GANs) frame generation as a competitive process between generator and discriminator networks [39], while diffusion models iteratively transform noise into coherent outputs through learned denoising procedures [40]. Large language models (LLMs), including GPT, Claude, and Gemini, extend these principles to language and multimodal reasoning by learning rich statistical representations from massive corpora of human-generated data [41–44].

These models provide a useful computational comparison to associative recombination, but the comparison should not be taken literally. Human episodic simulation involves scene construction, temporal projection, autobiographical memory, and self-referential processing supported by interactions among the default mode network, hippocampus, and other cortical systems [8,9]. Contemporary generative models instead rely on statistical

sequence modeling and learned representations. What the two cases share is the functional problem of producing new possibilities from previously acquired information, not a common mechanism for doing so.

However, generation alone does not guarantee creativity. Novel outputs must still be evaluated, refined, and situated within broader goals and constraints.

4.2. Evaluative Models

The central challenge of evaluation is not producing ideas but determining which ideas are worth pursuing. In humans, evaluative processes involve cognitive control, goal maintenance, and the refinement of candidate solutions. In artificial systems, analogous functions arise through mechanisms that impose constraints, optimize objectives, and assess the quality of generated outputs.

Neuro-symbolic systems represent one important class of evaluative models. By integrating neural representations with symbolic reasoning, these approaches combine the flexibility of statistical learning with the structure and interpretability of formal logic [45]. Automated planning, program synthesis, and search-based methods similarly provide mechanisms for evaluating large spaces of candidate solutions while enforcing task-specific objectives and constraints [46–48]. More recently, reasoning-oriented AI systems have increasingly incorporated iterative self-evaluation, reflection, and critique mechanisms to improve the quality of generated outputs [49,50].

These approaches address an evaluative problem similar to that faced during human creative cognition: selecting and refining promising solutions from a larger set of possibilities. Their implementation, however, differs substantially from biological executive control. Human evaluation draws on working memory, goal maintenance, cognitive control, and flexible adaptation mediated by distributed frontoparietal networks [12]; artificial systems typically rely on engineered objectives, search procedures, or learned optimization strategies.

Collectively, these approaches illustrate that creativity depends not only on the ability to generate possibilities but also on the capacity to identify which possibilities are useful, coherent, or worthy of further development. Evaluative processes therefore transform novelty into meaningful innovation.

4.3. Exploratory Models

Creativity often requires moving beyond familiar solutions in search of new possibilities. The central challenge of exploration is therefore deciding when and how to seek information that may have no immediate utility but could prove valuable in the future. In biological systems, this process is associated with curiosity, novelty seeking, and reward-guided learning. In artificial systems, it has been studied extensively within reinforcement learning.

A central problem in reinforcement learning is the exploration–exploitation dilemma: agents must balance exploiting known successful strategies against exploring potentially better alternatives [33]. One influential approach involves intrinsic motivation, in which agents are rewarded not only for achieving external goals but also for acquiring information, reducing uncertainty, or discovering novel states [51,52]. Curiosity-driven learning systems extend these ideas by using prediction errors, novelty signals, or information gain to guide exploration [53].

Meta-reinforcement learning extends these ideas by enabling agents to acquire strategies that support rapid adaptation across tasks rather than optimizing a fixed policy for a single environment [36,37]. This offers a computational counterpart to one aspect of biological exploration: using experience to decide when additional information is worth seeking. The underlying machinery is quite different. Dopaminergic and basal ganglia

circuits shape exploration through reward prediction errors, neuromodulation, and interactions with memory and executive systems [14,15], whereas reinforcement-learning algorithms generally rely on explicit objectives and mathematically specified update rules.

At the same time, current systems remain limited in their capacity for open-ended curiosity and autonomous information seeking, suggesting that important aspects of biological exploration remain to be captured computationally.

4.4. Coordination Models

Generation, evaluation, and exploration each contribute to creative behavior, but their relative importance is likely to vary across goals and contexts. Effective creative behavior may therefore depend not only on the capacity of each process, but also on how their engagement is regulated over time. Within GEEC, coordination refers to this regulatory problem: determining when to generate new ideas, when to refine existing possibilities, and when continued exploration may be useful. Unlike the generative, evaluative, and exploratory components of the GEEC framework, coordination is not proposed as a distinct cognitive faculty. Rather, it is conceptualized as a meta-process that regulates transitions among these components, allocating resources and shifting cognitive priorities as task demands change.

Attention mechanisms provide one important example of computational coordination, although coordination in modern AI increasingly extends beyond attention alone to include memory management, dynamic routing, planning, retrieval, and agent orchestration. Originally developed to selectively prioritize relevant information during sequence processing, attention has become a central component of transformer architectures and modern large language models [35]. Related approaches such as memory-augmented neural networks, retrieval-augmented generation, and mixture-of-experts architectures dynamically regulate which information or computational resources should be engaged for a given task [34,54,55].

These computational mechanisms illustrate several ways that coordination can be implemented without implying a neural analogue. In the brain, the salience network and related control systems contribute to attention, salience detection, and switching among distributed networks [27]. Artificial systems solve routing and control problems through learned policies, architectural constraints, retrieval, or explicit orchestration. The relevant comparison is therefore the regulatory problem itself: how to determine which information and which process should be engaged as goals and context change.

A central challenge for future AI systems is that coordination extends beyond selecting among existing computational modules. Biological coordination continuously integrates internal goals, environmental feedback, uncertainty, memory, and ongoing evaluation to regulate the balance among generation, evaluation, and exploration across multiple timescales. Current AI systems typically coordinate predefined components within relatively fixed architectures, whereas biological systems appear capable of dynamically reorganizing these interactions as tasks, goals, and contexts evolve. Developing computational mechanisms that exhibit this degree of adaptive coordination may therefore represent one of the major remaining challenges for machine creativity and, more broadly, for AGI.

Current coordination mechanisms provide important steps in this direction, but they remain limited in the flexibility with which they can reorganize behavior as goals, information, and task demands change.

4.5. Integrated Creative Architectures

The most direct computational test of GEEC comes from architectures that combine several of these functions rather than optimizing them in isolation.

Early cognitive architectures such as Copycat modeled creativity through interactions among perception, memory, attention, and analogy-making processes [56]. The Learning Intelligent Distribution Agent (LIDA) architecture similarly sought to integrate attention, memory, action selection, and learning within a cognitively inspired framework [57]. Thaler's Creativity Machine paradigm proposed that novel associations could emerge through perturbation and recombination within artificial neural networks [58]. This line of work was later extended in DABUS, which Thaler describes as generating novel concepts through dynamically forming associations among distributed neural representations [59].

Contemporary agent-based architectures increasingly combine language models with planning modules, memory systems, retrieval mechanisms, tool use, and external feedback [60]. Multi-agent frameworks further distribute cognitive functions across specialized agents that collaborate, critique, and refine one another's outputs [61]. These architectures are especially relevant to GEEC because they make it possible to ask whether integrating specialized functions changes creative performance in ways that cannot be explained by generation alone.

Current systems are still far from establishing that such integration is sufficient for human-like creativity. They do, however, provide increasingly tractable testbeds for asking how generation, evaluation, exploration, and coordination interact, and whether different configurations produce different forms of creative behavior.

5. Measuring Creativity in Artificial Systems

Questions about machine creativity are inseparable from broader debates concerning how creativity should be defined and measured. In psychology and creativity research, creativity is most commonly defined as the production of ideas or artifacts that are both novel and useful, although additional dimensions such as surprise, originality, aesthetics, and social value have also been proposed [1,62]. More recently, systems perspectives have emphasized creativity as an emergent consequence of interactions among cognitive, social, and environmental factors rather than as a single measurable trait [63,64]. These perspectives raise an important question for artificial intelligence: if creativity is multifaceted in humans, should its evaluation in artificial systems also be similarly multicomponent?

The GEEC framework suggests that the answer is yes. Within GEEC, creativity is understood as emerging from interactions among generative, evaluative, and exploratory processes that are dynamically regulated through coordination. Consequently, creativity should be assessed at multiple complementary levels rather than through a single aggregate score. Specifically, GEEC distinguishes *process engagement* (whether a particular creative process is recruited), *process effectiveness* (how successfully that process performs its intended function), and *creative outcomes* (the quality of the resulting products or behaviors). These levels should not be assumed to be equivalent. For example, a system may engage in extensive exploration without producing useful outputs, while another may achieve strong creative performance through relatively modest generation coupled with highly effective evaluation. Likewise, biological or artificial systems exhibiting similar creative outcomes may rely on fundamentally different balances among generation, evaluation, exploration, and coordination. Table 2 summarizes representative measures across these complementary levels.

Within this framework, generative creativity concerns the construction of novel ideas or representations. In humans, this component is commonly assessed through divergent thinking tasks, fluency measures, and evaluations of originality [2,65]. In artificial systems, process-level measures include representation diversity, latent-space coverage, or sampling diversity, whereas outcome-level measures include novelty, surprise, and originality relative to previous outputs or training data. Measures of representational information and

compressibility, which have been used to characterize the structure and efficiency of high-dimensional neural representations [66], may offer an additional approach for examining the internal representations that support generative behavior. Whether such measures are informative about creativity itself, however, remains an open question. Novelty alone is likewise insufficient: highly unusual outputs are not necessarily creative if they lack utility or relevance.

Table 2. Illustrative measures for assessing the components of the GEEC framework. For purposes of operationalization, process-level measures can capture both process engagement (whether a component is recruited) and process effectiveness (how appropriately or successfully it operates), whereas outcome-level measures characterize the products or behaviors that result from those processes. The AI measures shown are intended as candidate operationalizations rather than established or validated creativity benchmarks.

GEEC Component	Representative Neural Systems	Representative Human Measures	Illustrative AI Measures
Generative	Default Mode Network (DMN), Hippocampus	Divergent thinking, fluency, originality	**Process:** representation diversity, latent-space coverage, sampling diversity **Outcome:** novelty, originality, surprise
Evaluative	Executive Control Network (ECN)	Usefulness, expert ratings, creative achievement	**Process:** self-evaluation, iterative refinement, planning, preference modeling **Outcome:** coherence, goal attainment, task performance, human preference ratings
Exploratory	Dopaminergic systems, Basal Ganglia, Hippocampus	Curiosity, openness to experience, novelty seeking	**Process:** information gain, intrinsic reward, curiosity-driven behavior, state-space coverage **Outcome:** discovery of novel solutions, adaptation to unfamiliar environments
Coordination	Salience Network (SN), Frontoparietal Control Systems	Cognitive flexibility, task switching	**Process:** switching timing, context-sensitive strategy selection, transition appropriateness, dynamic resource allocation, module orchestration **Outcome:** stable performance under changing goals, efficient coordination across modules, rapid adaptation to changing goals and task demands

Evaluative creativity concerns the ability to assess, refine, and select candidate ideas. Human creativity research has long relied on expert judgments, measures of usefulness, and assessments of creative achievement to evaluate the quality of creative products [62]. Amabile's Consensual Assessment Technique (CAT) provides a complementary product-based approach in which creativity is evaluated through the independent judgments of individuals with relevant domain expertise [67]. Because it does not impose a universal scoring rule, the CAT is particularly well suited to assessing contextualized, real-world creative outputs and may offer a useful framework for evaluating artifacts produced by artificial systems. In artificial systems, process-level measures include self-evaluation, iterative refinement, planning, or preference modeling, whereas outcome-level measures include coherence, goal attainment, task performance, and human preference ratings.

More broadly, recent benchmarks for large language models have increasingly emphasized multidimensional evaluation spanning reasoning, factuality, instruction following, planning, and alignment with human preferences rather than relying on a single aggregate benchmark [68,69]. Although these benchmarks were not designed specifically to assess creativity, they reinforce a broader trend toward evaluating complex AI systems through multiple complementary dimensions. From the perspective of GEEC, creativity represents another example of this principle: meaningful assessment requires evaluating not only creative outcomes, but also the processes that generate, evaluate, explore, and coordinate ideas.

Exploratory creativity reflects the search for novel information, experiences, or solution strategies. In humans, exploratory tendencies are associated with curiosity, openness to experience, novelty seeking, and intrinsic motivation [63]. In artificial systems, process-level measures include information gain, intrinsic rewards, curiosity-driven behavior, and state-space coverage, whereas outcome-level measures include successful adaptation to unfamiliar environments or discovery of previously unseen solutions [51–53]. From this perspective, creativity depends not only on generating ideas but also on expanding the space of possibilities from which those ideas emerge.

Coordination concerns the regulation of interactions among generative, evaluative, and exploratory processes rather than representing a separate source of creativity itself. Unlike the other GEEC components, coordination is not most naturally assessed through the quality of creative outputs. Instead, it should be evaluated in terms of how effectively a system regulates interactions among the other creative processes across changing goals and contexts. Process-level measures may therefore include switching timing, transition appropriateness, context-sensitive strategy selection, dynamic resource allocation, routing efficiency, or module orchestration. Outcome-level measures include stable creative performance under changing goals, efficient integration of specialized modules, robustness to changing task demands, and rapid adaptation to novel contexts. Although these candidate measures provide a starting point for operationalizing coordination, GEEC currently provides a functional definition and candidate operational measures rather than a validated measurement model of the construct. Developing reliable measures of coordination therefore represents an important direction for future research as increasingly sophisticated AI systems begin to integrate multiple interacting creative processes.

Taken together, these considerations suggest that creativity is best understood as a multicomponent construct rather than a single score or capability. Rather than asking whether an artificial system is creative in a binary sense, it may be more informative to ask how effectively it generates novel representations, evaluates candidate ideas, explores uncertain possibilities, and coordinates these processes over time. This perspective also suggests that different creative domains may rely on different balances among these processes, implying that no single benchmark is likely to capture the full breadth of creative intelligence. Instead, creativity should be viewed as a dynamic and context-dependent phenomenon emerging from interactions among multiple complementary processes.

6. Predictions of the GEEC Framework

The GEEC framework is intended not only as an organizing perspective on creativity, but also as a source of empirically testable hypotheses spanning neuroscience, cognitive psychology, and artificial intelligence. Because GEEC is formulated as a functional framework rather than a mechanistic model, these predictions concern relationships among cognitive processes and computational principles rather than one-to-one mappings between specific brain regions and AI architectures. If GEEC captures essential features of creative cognition, several predictions follow (Table 3).

Table 3. Illustrative empirical predictions generated by the GEEC framework. Rather than prescribing specific experiments, these examples demonstrate how the framework may be evaluated across neuroscience and artificial intelligence.

GEEC Prediction	Example Neuroscience Test	Example AI Evaluation	Evidence Supporting GEEC
Creative behavior depends on separable functional processes	Selectively perturb generative, evaluative, exploratory, or coordination processes using lesion studies, non-invasive brain stimulation, pharmacological manipulation, or patient populations.	Perform targeted ablation of generation, evaluation, exploration, or coordination mechanisms within creative AI systems.	Different perturbations produce distinct profiles of creative strengths and weaknesses rather than a uniform reduction in creativity.
Creativity emerges from coordinated interactions	Compare models based on dynamic functional connectivity, network flexibility, or state transitions with models based only on activity within individual brain networks.	Compare generation-only models with integrated architectures that explicitly regulate generation, evaluation, and exploration.	Interaction-based models explain creative performance better than models considering each process independently.
Process-level assessment outperforms single-score evaluation	Measure behavioral and neural indicators of generation, evaluation, exploration, and coordination separately rather than relying solely on overall creativity scores.	Develop benchmarks that independently quantify novelty, usefulness, exploration, and coordination before relating them to overall creative performance.	Multicomponent assessments explain individual differences and system behavior better than aggregate creativity scores alone.

6.1. Prediction 1: Creative Behavior Depends on Separable Functional Processes

GEEC predicts that generation, evaluation, and exploration contribute distinct functional roles to creative behavior and therefore should produce dissociable patterns of impairment or enhancement when selectively manipulated. Deficits in generation should primarily reduce the fluency, diversity, or originality of candidate ideas. Deficits in evaluation should impair the ability to identify useful, coherent, or contextually appropriate solutions despite relatively preserved idea generation. Deficits in exploration should reduce curiosity, information seeking, and the discovery of alternative solution strategies. In contrast, disruptions to coordination should not necessarily impair any individual process, but should alter when and how those processes are engaged. Such disruptions may be reflected in poorly timed or contextually inappropriate transitions between generation, evaluation, and exploration—for example, evaluating ideas prematurely, continuing to explore after additional search is no longer useful, or failing to shift from generation to evaluation as task demands change.

This prediction is directly testable across both biological and artificial systems. In neuroscience, selective perturbations through lesion studies, non-invasive brain stimulation, pharmacological manipulation, or neurological disorders could determine whether disruption of different neural systems produces distinct creative deficits. In artificial intelligence, analogous experiments could selectively ablate or constrain generative, evaluative, exploratory, or coordination mechanisms to determine whether each manipulation produces the predicted pattern of behavioral changes.

6.2. Prediction 2: Creativity Emerges from Interactions Among Processes

Although generation, evaluation, and exploration each contribute to creative behavior, GEEC predicts that creativity will be explained more accurately by their coordinated interaction than by any individual process alone. In neuroscience, this implies that measures of dynamic functional connectivity, network flexibility, and transitions among large-scale brain networks should explain creative performance beyond activity within individual networks. Likewise, creative behavior should depend on how exploratory experiences

influence subsequent generation and how evaluative processes iteratively refine candidate ideas.

GEEC further predicts that increasing generative, evaluative, or exploratory capacity independently will not necessarily improve creative performance. Instead, the benefit of increasing any one process should depend on whether coordination appropriately regulates when and to what extent that process is engaged. For example, increasing exploration should improve performance when additional information seeking is useful, but may impair performance when continued exploration delays evaluation or commitment to a solution.

A similar prediction applies to artificial intelligence. Systems capable of dynamically coordinating generative, evaluative, and exploratory capabilities should exhibit greater adaptability, robustness, and transfer than systems emphasizing generation alone. This hypothesis can be examined through ablation studies comparing integrated agent architectures with purely generative models, as well as by determining whether explicit coordination mechanisms improve performance under changing objectives or unfamiliar tasks. Conversely, evidence that creative performance is fully explained by independent generative, evaluative, and exploratory capacities, with no incremental contribution from their context-dependent regulation, would argue against coordination as a necessary component of the framework.

6.3. Prediction 3: Creativity Should Be Assessed as a Multicomponent Process

GEEC predicts that creativity cannot be adequately characterized by a single benchmark or summary score. Instead, assessment should distinguish among generative capacity, evaluative ability, exploratory behavior, and the effectiveness with which these processes are coordinated. Importantly, process engagement, process effectiveness, and final creative outcomes should not be assumed to be equivalent. GEEC further predicts that creative performance should depend on the context-sensitive balance among generation, evaluation, and exploration and on the effectiveness with which that balance is coordinated, such that different process-level configurations may produce similar levels of overall creative performance. For example, broad exploration may increase the diversity of candidate ideas without improving their eventual usefulness, whereas highly effective evaluation may improve final performance despite relatively modest generative output.

This perspective suggests that future creativity benchmarks should explicitly quantify multiple components rather than relying solely on overall task performance or product quality. In neuroscience, this may involve combining behavioral measures with neural indicators of generation, evaluation, exploration, and coordination. In artificial intelligence, multidimensional benchmarks could independently quantify novelty, usefulness, exploratory behavior, and coordination effectiveness, allowing systems with similar overall creativity scores to be distinguished according to the computational strategies they employ.

Collectively, these predictions move GEEC beyond a descriptive taxonomy by providing hypotheses that can be evaluated experimentally across neuroscience and artificial intelligence. At the same time, these predictions represent initial hypotheses derived from the framework rather than an exhaustive set of uniquely diagnostic tests. Stronger tests of GEEC will require direct comparisons with competing accounts of creative behavior, particularly to determine whether measures of process interaction and coordination provide explanatory or predictive value beyond generative, evaluative, and exploratory capacities considered independently. More broadly, the framework suggests that progress toward understanding and engineering creative intelligence will depend not simply on improving individual capabilities, but on understanding how multiple complementary

processes are dynamically coordinated to produce flexible, adaptive, and context-sensitive creative behavior.

Beyond these specific predictions, GEEC predicts that successful creativity will not correspond to a single optimal process profile. Instead, the balance among generation, evaluation, and exploration, and the coordination of those processes, should vary according to the goals, constraints, and demands of a particular creative domain. Domains emphasizing relatively unconstrained novelty, such as early-stage artistic ideation, should place greater demands on generation and exploration, whereas domains requiring solutions under strong external constraints, such as scientific or engineering problem solving, should place relatively greater demands on evaluation and coordination. Importantly, GEEC predicts that these differences should be observable at the process level even when overall creativity ratings are matched. Likewise, artificial systems optimized for different creative domains may exhibit different balances among these processes while achieving comparable levels of overall creative performance. These predictions suggest that creativity should not be viewed as a single latent capability, but rather as a family of adaptive behaviors whose underlying functional organization depends on context, goals, and domain-specific constraints.

Rather than asking whether a system is "creative," the GEEC framework therefore suggests a more informative question: *what balance of generative, evaluative, and exploratory processes, and what form of coordination among them, enables creativity for a particular domain or objective?*

7. Discussion

Creativity remains a challenging phenomenon to explain in both biological and artificial intelligence. Although recent advances in AI have produced systems capable of generating increasingly sophisticated text, images, music, and code, fundamental questions remain regarding the mechanisms that underlie creative behavior and the extent to which machine creativity resembles human creativity. This review has argued that neuroscience provides a useful perspective on these questions by suggesting that creativity emerges not from a single cognitive faculty, but from interactions among multiple processes.

The GEEC framework formalizes this perspective by proposing three primary processes dynamically regulated by coordination that appear repeatedly across neuroscience, cognitive science, and artificial intelligence. Generative processes construct novel representations and possibilities. Evaluative processes assess those possibilities relative to goals and constraints. Exploratory processes drive curiosity, information seeking, and adaptation to uncertainty. Coordination mechanisms regulate interactions among these processes and enable flexible behavior across contexts and timescales. Although these components do not map one to one onto specific brain regions or AI architectures, they provide a common language for describing creativity across biological and artificial systems.

The processes described by the GEEC framework should not be interpreted as equivalent psychological constructs. Generation, evaluation, and exploration describe core functional processes involved in creative cognition, whereas coordination refers to the higher-order regulation of interactions among those processes. This distinction helps reconcile GEEC with existing cognitive and network-neuroscience models while emphasizing the importance of dynamic control in creative behavior.

One implication is that creativity should not be identified solely with generative ability. Nor does GEEC treat creativity as a necessary or sufficient criterion for general intelligence; rather, creativity provides a tractable domain in which to examine capacities relevant to flexible, adaptive behavior. Contemporary discussions of AI creativity frequently emphasize the remarkable generative capabilities of large language models and diffusion models. However, human creativity also involves the ability to evaluate ideas, seek new

information, and dynamically coordinate these processes, with their relative contributions likely varying across contexts and domains. From this perspective, current AI systems may be viewed as implementing different components of creative cognition to varying degrees rather than possessing creativity as a singular capability. This perspective also suggests that future advances in machine creativity may arise less from improvements to generative models alone and more from increasingly sophisticated interactions among generative, evaluative, and exploratory mechanisms through effective coordination.

The framework also has important limitations. The proposed components are intended as functional abstractions rather than exhaustive descriptions of creativity, and the boundaries between them are necessarily imprecise. More fundamentally, this proposed organization of creative cognition remains a theoretical decomposition rather than an empirically established architecture. Although each component is supported by the existing cognitive and neuroscientific literature, it remains to be determined whether these processes are dissociable in the manner proposed by GEEC, whether alternative decompositions provide equal or greater explanatory power, and whether the framework generalizes across different forms and domains of creativity. Creativity is also influenced by developmental, social, cultural, and environmental factors that are only partially captured by the present framework. In particular, GEEC is primarily a within-system account of the functional processes supporting creative cognition and does not attempt to explain how creative value is socially constructed or historically situated. Sociocultural accounts emphasize that whether an idea or product is recognized as creative depends partly on communities, institutions, cultural norms, and the environments in which creative activity occurs [63,64]. Creativity may also emerge through collaborative or distributed processes that cannot be straightforwardly reduced to processes within a single individual or artificial system. These phenomena therefore represent an important boundary of the present framework rather than additional processes that GEEC is intended to subsume. Furthermore, many of the computational analogies discussed throughout this review remain speculative and should be understood as heuristic functional correspondences rather than evidence of shared underlying mechanisms. Similarities between biological and artificial systems do not imply shared mechanisms, and neuroscience should not be viewed as providing a direct blueprint for AGI.

These limitations make empirical tests of the framework especially important. Neuroscientifically, GEEC predicts that creative performance should depend on dynamic interactions among neural processes supporting generation, evaluation, and exploration, regulated through higher-order coordination mechanisms rather than activity within any single system. Computationally, GEEC predicts that artificial systems integrating these functions through effective coordination may exhibit richer and more adaptive creative behavior than systems emphasizing any single component. Importantly, increasing generative, evaluative, or exploratory capacity alone should not necessarily improve creative performance; its benefit should depend on how that capacity is coordinated with other processes and on the demands of the task. Different creative domains may therefore favor different process-level configurations, and similar levels of overall creative performance may arise from distinct underlying profiles. These predictions provide concrete opportunities to test not only whether the individual components contribute to creativity, but whether their proposed functional relationships offer explanatory value beyond existing accounts. Future work should determine whether these predicted relationships generalize across biological and artificial creative systems.

GEEC also illustrates the potential value of sustained exchange between neuroscience and artificial intelligence. Throughout the history of artificial intelligence, neuroscience and computation have repeatedly informed one another, from early neural networks to reinforce-

ment learning and attention-based architectures. Creativity may represent another domain in which this exchange is particularly informative. By identifying functional principles that can be examined across brains and machines, neuroscience can provide conceptual tools and hypotheses for understanding emerging AI systems, while advances in AI may offer new computational models and experimental testbeds for investigating mechanisms underlying human creativity. Such comparisons should remain at the level of functional and computational principles unless stronger evidence supports direct mechanistic correspondence.

Ultimately, the question of whether artificial systems are "truly" creative may be less informative than understanding the processes through which creative behavior emerges. GEEC proposes that creativity can be examined as an emergent consequence of interactions among complementary generative, evaluative, and exploratory processes whose engagement is dynamically regulated by coordination and whose relative contributions vary across contexts and goals. Creativity thus provides a concrete setting in which to investigate how intelligent systems generate possibilities, evaluate them against constraints, explore uncertain solution spaces, and regulate these activities as circumstances change. For artificial intelligence, the challenge may therefore extend beyond building increasingly powerful generators to understanding whether and how systems can balance these complementary functions in flexible and context-sensitive ways. The ultimate value of GEEC will not depend on whether every proposed mapping proves correct, but on whether the framework generates productive hypotheses that distinguish among competing accounts of creativity and motivate new experiments across neuroscience and artificial intelligence.

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